GRIFFITH COLLEGE

Assignment Cover Sheet

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Faculty: _____Computing_____

Programme: _____BSCO/H_____ Stage/Year: _____3_____

Module:_____WT_____

Study Mode: Full Time _____ Part-time _____

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Assignment Title: _____Assignment Four_____

Additional Relevant Information (e.g. number of pieces submitted etc.):

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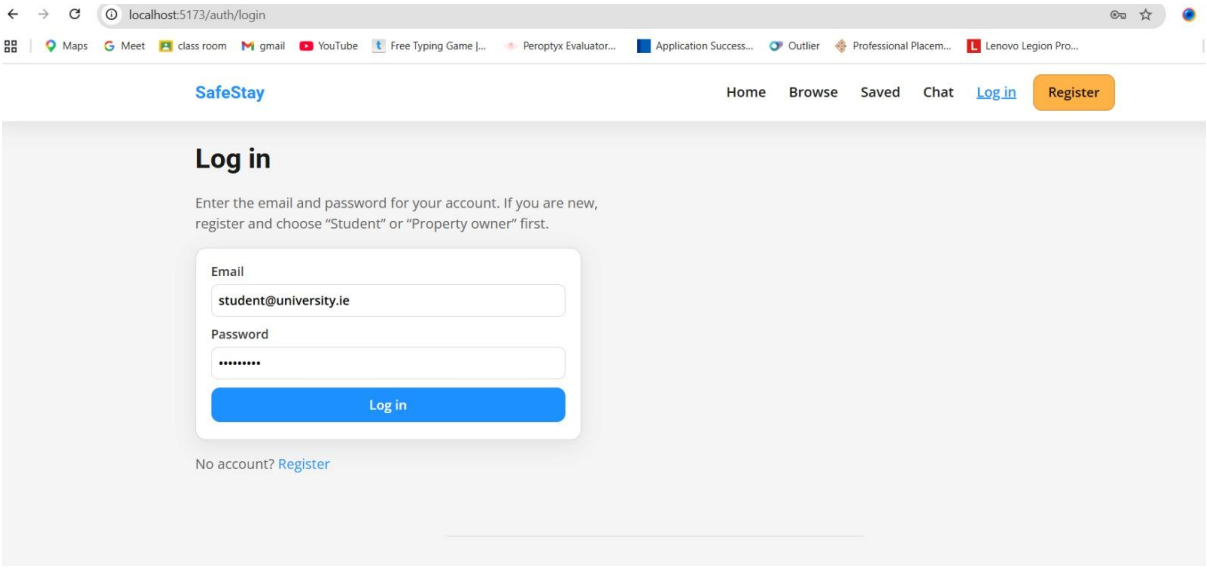
Project Description

SafeStay is a full-stack student accommodation platform developed to help students discover safe and reliable rental properties while allowing property owners to manage accommodation listings efficiently. The project was built using a React.js single-page application with Vite on the frontend, connected to an Express.js REST API and MongoDB database on the backend. The system also supports a local mock mode using browser storage, allowing the interface to function without a backend connection during demonstrations or offline testing.

The application allows users to browse accommodation listings using keyword search, property-type filters, and monthly price filtering with validation handling for invalid ranges. Each listing displays rental information, descriptions, uploaded images, landlord details, and verification indicators where available. Students can save favourite listings and contact property owners through a listing-based messaging system. In local development, the React app is paired with the Express API using the Vite dev proxy (/api → localhost), which keeps messaging session-stable and fully functional on localhost—students and owners exchange messages against the MongoDB-backed API without cross-origin complications.

Authentication functionality includes user registration, login, logout, role-based behaviour for students and owners, and session persistence using API-based authentication. Property owners can create, view, and delete their own listings through the frontend interface, while listing updates are supported at API level.

The project demonstrates frontend and backend integration, REST API development, session-based authentication, role-aware navigation and access checks, data operations spanning create/read/update/delete on the backend with create/read/delete surfaced in the SPA (listing edit via API while the edit screen is deferred), responsive user interface design, and deployment using Render and MongoDB Atlas within a MERN-style architecture.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Lenovo Legion Pro...
SafeStay
Home
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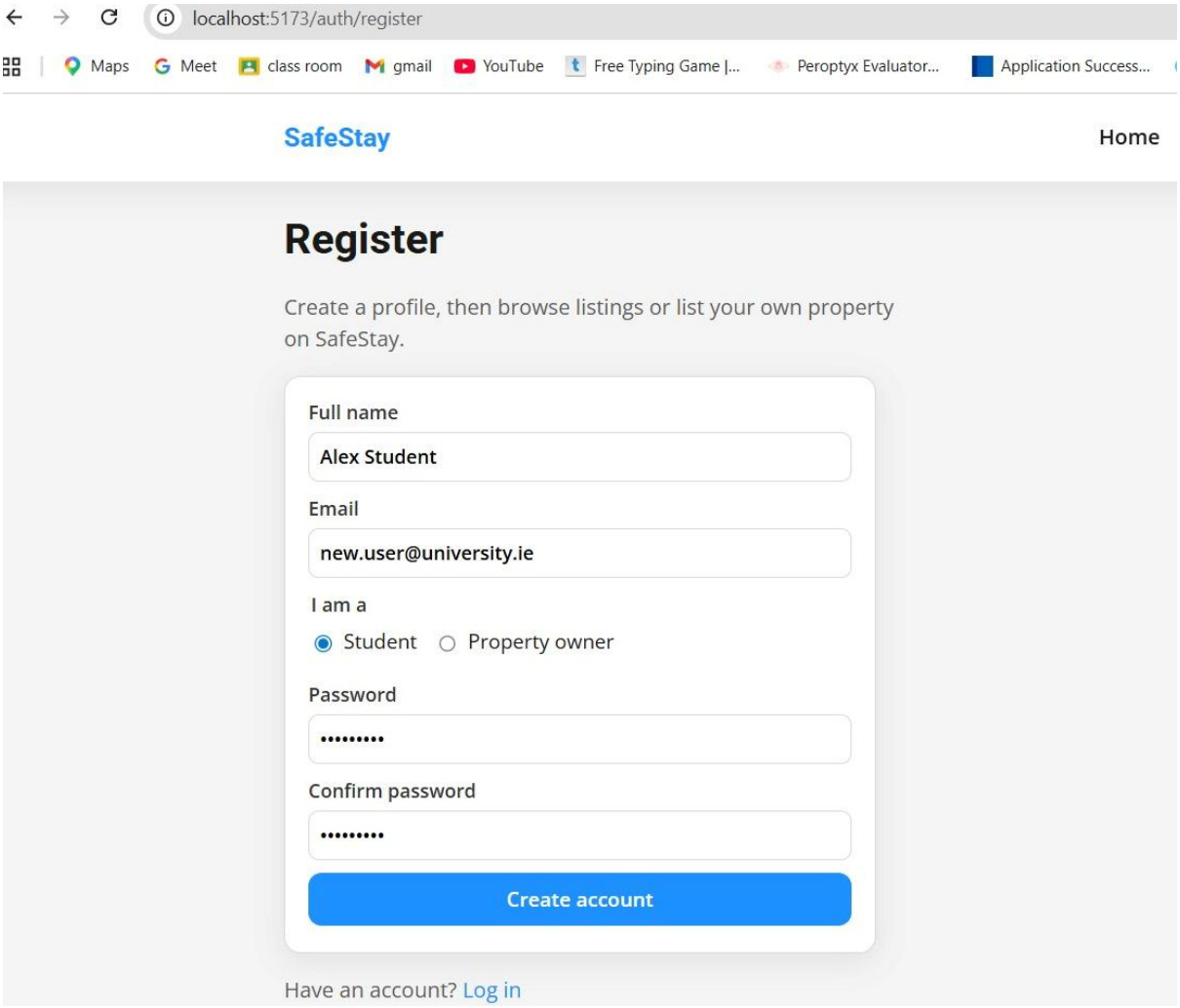
Chat
Log in
Register
Log in
Enter the email and password for your account. If you are new, register and choose "Student" or "Property owner" first.
Email
student@university.ie
Password
Log in
No account? Register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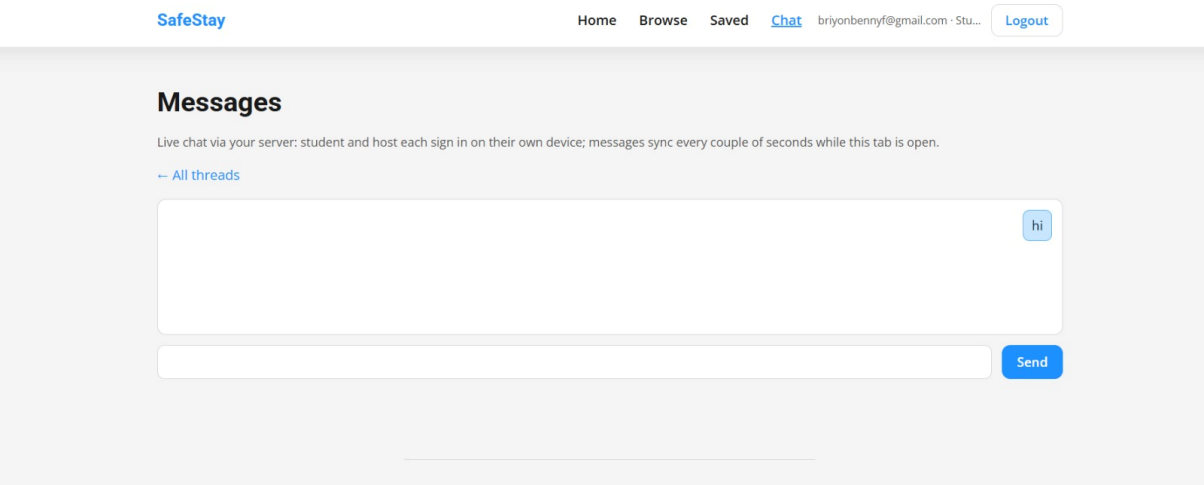
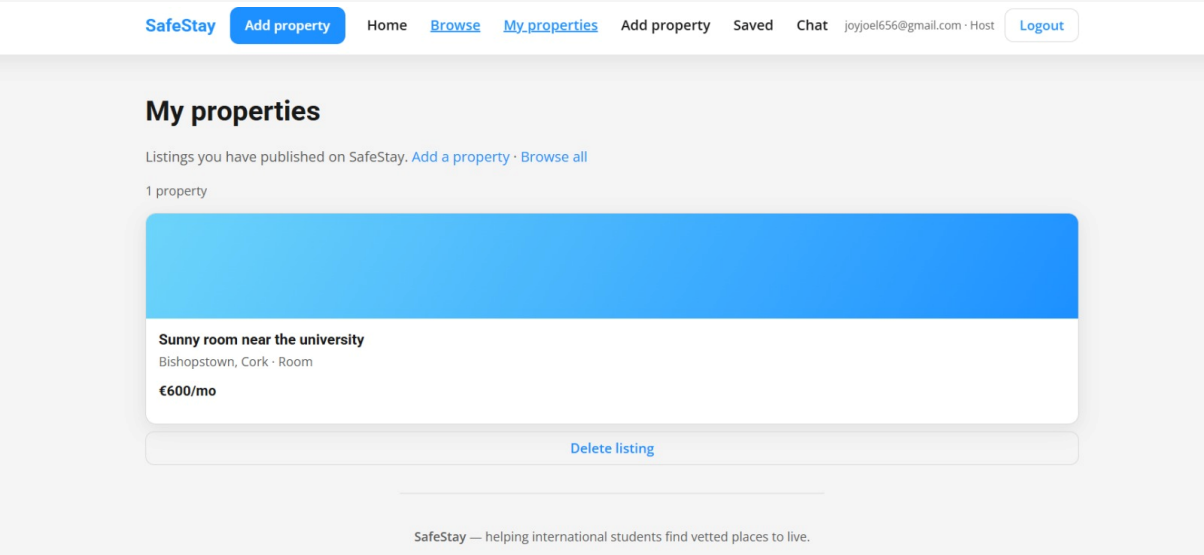
Application Success...
SafeStay
Home
Register
Create a profile, then browse listings or list your own property on SafeStay.
Full name
Alex Student
Email
new.user@university.ie
I am a
Student
Property owner
Password
Confirm password
Create account
Have an account? Log in





Working Functionalities

The SafeStay platform includes several functional frontend and backend features that support student accommodation browsing and property management.

Users can register and log in securely using role-based authentication for students and property owners. Session persistence is handled through the backend API using session cookies, allowing users to remain authenticated across refreshes. Navigation and access behaviour change dynamically depending on the logged-in user role.

The listings system allows owners to create, view, and delete their own property listings through the React interface. Listing updates are supported by the backend API, although an edit listing screen is not currently exposed in the SPA. All users can browse listings using keyword search, property-type filters, and price filtering with validation messages for invalid ranges.

Students can save and remove favourite listings with synchronisation between the frontend and backend when API mode is enabled.

The platform includes a messaging system that supports listing-specific conversations between students and property owners. On localhost, messages send and load reliably through the REST API with thread listing, conversation history, and polling-based refresh in the inbox and chat views. When the frontend and API are hosted on different Render URLs, browsers treat them as separate sites, so CLIENT_ORIGIN must match the SPA origin so session cookies attach to authenticated requests—a deployment configuration step, not a limitation of chat logic locally.

Additional implemented features include responsive layouts, local mock mode using browser storage, reporting acknowledgement functionality, shared validation handling, API error responses, and deployment-ready REST APIs for testing and integration.

Project Development and Improvements

Since the original proposal in Assignment 1, the SafeStay project developed significantly in both functionality and technical implementation. The initial proposal focused mainly on a simple accommodation platform with authentication, listings, favourites, and communication features. During development, the project expanded into a more complete MERN-style application with integrated frontend and backend systems.

Several improvements were implemented beyond the original scope. The frontend was redesigned using React.js with reusable layouts, React Router navigation, responsive UI styling, and shared state management using React Context. Additional validation was added for login, registration, listing creation, and filtering forms to improve usability and reduce invalid input handling. A local mock mode was also introduced using browser storage, allowing the application to operate without a backend connection during demonstrations and testing.

Messaging was implemented as listing-scoped REST endpoints backed by MongoDB messages, supporting conversation loading, sending, inbox/thread retrieval, and polling. During local testing, behaviour was dependable end-to-end because the SPA and API share a simple same-machine / proxied development setup. Packaging the same behaviour for Render required attention to cross-origin session cookies—in particular setting CLIENT_ORIGIN on the API and ensuring the static site was built with the correct VITE_API_BASE_URL—so production matches the smooth experience already proven on localhost.

Some planned features were simplified due to time and technical constraints. Although the backend API supports updating listings, a complete owner edit-listing interface was not implemented in the React frontend. Real-time communication using Socket.IO or WebSockets was also not implemented, with polling-based refresh used instead. The

reporting functionality currently provides acknowledgement feedback without a complete moderation dashboard or persistent admin workflow.

Given more time, the team would like to implement real-time messaging, notification systems, a complete owner edit-listing interface, admin moderation tools, cloud image storage integration, and additional accessibility improvements.

Team Contributions

Briyon Benny (3144036):

- Listings pages and listing detail views
- Owner/student role-aware behaviour
- Listing filtering and property management
- CRUD functionality for listings
- Documentation support

Joyal Joy (3147134):

- Authentication system
- Login/Register forms
- Session handling and validation
- API authentication integration
- Error handling implementation

Ayo-oluwa Joshua Olajide (3139178):

- Shared layouts and frontend integration
- Favourites and messaging interface
- API helper integration
- Deployment setup and testing
- Documentation and Render deployment.

Technologies Used

- React.js
- Vite
- React Router
- JavaScript
- HTML5
- CSS3
- Node.js
- Express.js
- MongoDB

- Mongoose
- Express Session
- Connect-Mongo
- Multer
- Render.com
- Bruno API Testing

Coding Standards

The project follows consistent coding standards across both frontend and backend components. Clear and understandable identifier names were used throughout the application to improve readability and maintainability. Consistent indentation and formatting practices were followed in React components, Express routes, middleware, and database models.

Comments were added to non-trivial sections of code, particularly around API routes, authentication logic, validation handling, and shared utility functions. The application structure was separated into client and API directories to maintain clear organisation and separation of concerns.

Deployment Links

Frontend (Render):

https://safestay-frontend.onrender.com

Backend/API (Render):

https://safestay-app-e458.onrender.com

Conclusion

The SafeStay project successfully delivers a modern student accommodation platform using integrated frontend and backend technologies. The application demonstrates authentication systems, listing management, favourites functionality, messaging support, responsive design, REST APIs, validation handling, deployment support, and role-aware functionality for students and property owners.

The project reflects significant progress since the original proposal and provides a strong foundation for future improvements. Messaging is fully exercised against the API on localhost; hosted operation depends on the same session and CORS settings described in the deployment guide. While some planned enhancements—such as real-time messaging, admin moderation tools, and a dedicated owner edit-listing screen in

the SPA—were deferred, the current build demonstrates a functional MERN-style stack suitable for student accommodation browsing, enquiries, and deployment.

Resources

- https://react.dev/
- https://expressjs.com/
- https://www.mongodb.com/docs/
- https://mongoosejs.com/
- https://vitejs.dev/
- https://render.com/docs
- https://stackoverflow.com/
- https://developer.mozilla.org/